

Subtle Gaze Guidance for 360° Content by Gradual Brightness Modulation and Termination of Modulation by Gaze Approaching

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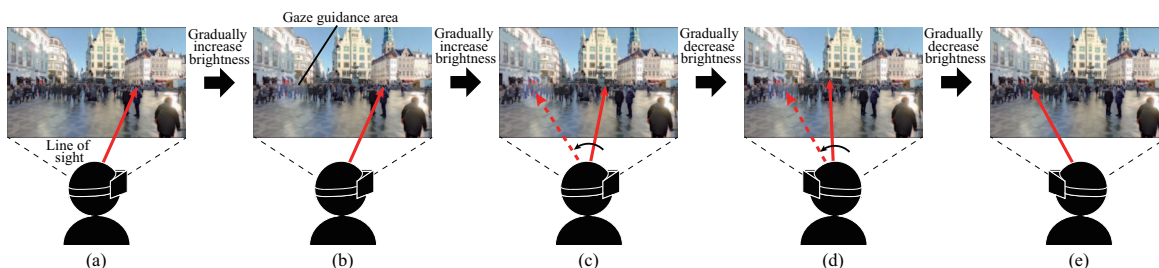


Figure 1: We propose a subtle gaze guidance method for 360° content that combines two techniques; gradual brightness modulation and termination of modulation by gaze approaching. (a) Original 360° content. (b) The brightness of the gaze guidance area ($r = 6^\circ$) is gradually increased. (c) The brightness will be maintained when it reaches a threshold that was determined so that it is perceivable even in the peripheral vision. (d) As the user's gaze approaches ($< 20^\circ$) the guidance area, the brightness starts to decrease gradually. (e) Its brightness has already returned to the original value when the user looks at the gaze guidance area.

ABSTRACT

On VR, users do not always see the specific contents that the creators want them to focus on. For the creators to provide users with the immersive experience they intend, it is necessary to naturally guide the user's gaze to relevant spots in the virtual space. In this paper, we propose a subtle gaze guidance method for 360° content combining two techniques; gradual brightness modulation and termination of modulation by gaze approaching the guidance area. The experimental results show that our method significantly contributes to a more natural and less disturbing viewing experience while maintaining relatively high guidance performance.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Computing methodologies—Computer graphics—Graphics systems and interfaces—Perception

1 SUBTLE GAZE GUIDANCE METHOD

By using a head-mounted display (HMD), virtual reality (VR) content creators can provide users with an immersive experience, which is not possible with traditional films and plays. On the other hand, users do not always see the specific events and gimmicks on the VR content that the creators want them to focus on. For the creators to provide users with the VR experience they intend, it is necessary to naturally guide the user gaze in the direction they wish to show in the virtual space.

In this paper, we propose a subtle gaze guidance method that does not interfere with a user experience, which can be easily introduced to any 360° content by post-processing. Our objective is to

develop a general-purpose gaze guidance method by applying inconspicuous modulation to reduce uncomfortable sensations of the user in gaze guidance, independently of the content.

Our method combines two techniques; gradual brightness modulation and termination of modulation by gaze approaching the guidance area [1, 2]. The process flow of the gaze guidance in the proposed method is illustrated in Fig. 1¹. It is known that the more attention is drawn to the area with higher brightness [3], and the proposed method is expected to reduce irritation and discomfort of the user by gradual modulation compared to sudden modulation. Furthermore, when the user's gaze approaches the guidance area, the modulation starts to decrease gradually back toward the original state to prevent the modulation from remaining in the user's consciousness. All the parameters such as the coefficient of the gamma correction function were determined heuristically.

Many existing gaze guidance techniques require the content creator to introduce artificial visual stimuli such as an arrow into the content, or change the content by camera zooming or a character's pointing so that the gaze guidance fits the worldview and the story. On the other hand, it may be possible to guide the user gaze without interfering with the natural, immersive viewing experience by simply introducing our method to VR content as a post-processing technique. In addition, our method makes it difficult for the user to see the modulated content in the user's central field of view so that they will perceive the content almost intact.

2 EXPERIMENT

The user studies reported in the present paper were reviewed and approved by the institutional review board. All the participants provided their written informed consent to participate in the user studies. In the experiment, we compare the following four different gaze guidance conditions on the participants who are not aware of the intention of gaze guidance. The four conditions are named *Gradual-Continuation* (GC), *Gradual-Termination* (GT), *Sudden-Continuation* (SC), and *Sudden-Termination* (ST), respectively. *Gradual-** are conditions that the modulation appears and

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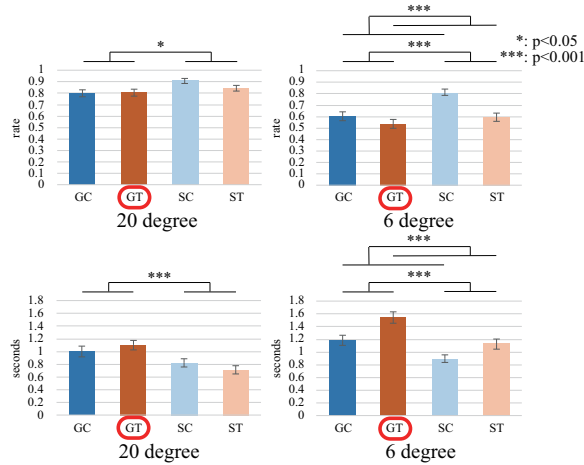


Figure 2: (Top) Success rate of the line of sight guided within a certain angle around the target. (Bottom) Average time for the line of sight to first move within a certain angle around the target.

disappears gradually. *Sudden* are conditions that the modulation appears and disappears instantly. *Continuation* are conditions that the guidance area remains modulated during gaze guidance. *Termination* are conditions that the modulation is terminated when the user gaze approaches the guidance area. *Gradual-Termination* (GT) condition is the proposed method.

In the experiment, the gaze-guiding task, each lasting between 25 to 35 seconds, is repeated 16 times for four 360° images for four gaze guidance conditions in a randomized order. Participants observe the 360° images with 3 DoF (Degrees of Freedom) head tracking (rotation only). Participants answer a questionnaire between the tasks. First, a small white dot is presented for three seconds in the frontal direction in black space. Then one of the four 360° images is presented. There is a no-guidance period with a random duration between one and three seconds before actually starting the gaze guidance. Then, gaze guidance begins by using one of the four gaze guidance conditions for four seconds. The center of the gaze guidance area is randomly positioned within a square region of 60°×60° in the frontal direction.

Their line of sight is recorded during the task and after each 360° image they are asked to answer the questionnaire Q1–Q5 (see Fig. 3), in a 7-point Likert scale (7 being the strong agreement and 1 being the strong disagreement). This is repeated four times for each of the 360° images for a particular gaze guidance condition. Then, the entire procedure is repeated for each of the four gaze guidance conditions. The orders of the 360° images and that of the gaze guidance conditions are randomized.

3 RESULTS AND DISCUSSION

Ten males and two females aged between 22 and 25 years participated in the study. Figure 2 shows the results of analysis of the participants' gaze during gaze guidance. And the results of the questionnaire are shown in Fig. 3. The proposed method (the GT condition) is indicated by a red circle in each figure. A two-way ANOVA was used to analyze the gaze data, and a two-way rank-transformed ANOVA [4] was used to analyze the questionnaire results. The results of the experiment confirmed that gaze guidance becomes significantly less uncomfortable with the gradual brightness modulation. It was also confirmed that the gradual gaze guidance is less effective than the sudden gaze guidance in terms of angular distance and speed. However, the line of sight moves within 20° from the center of modulation at least once for more than 80% of the trials

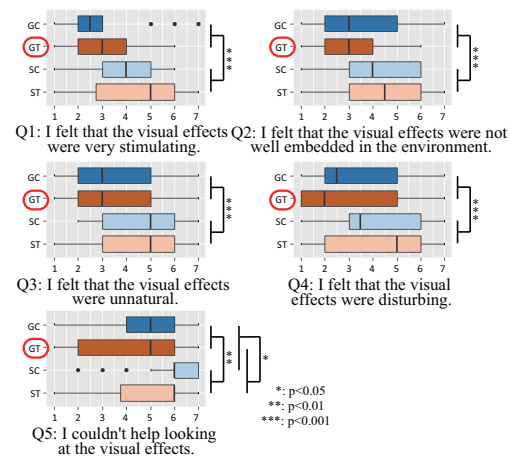


Figure 3: Questionnaire results.

in all of the guidance conditions. Overall, it can be said that the gradual brightness modulation is a moderately effective, unobtrusive gaze guidance technique.

On the other hand, according to the results of Q5 in the questionnaire, regarding the termination of modulation by gaze approaching, it was found that the participants felt more weakly that they could not help looking at the guidance area with this technique when they did not know the intention of gaze guidance and were allowed to look at any directions they preferred. We found no significant differences in the performance of rough gaze guidance (20°) and most of the subjective impressions for the unaware participants in the experiment. This means that they felt more freedom in their looking directions, whereas their gaze was actually guided to a comparable extent. The performance of precise gaze guidance (6°) was naturally lower with the termination of modulation by gaze approaching, however, it can be said that the termination of modulation by gaze approaching is helpful to make gaze guidance more unobtrusive. In the experiments, the gaze guidance area was randomly positioned within which there were no meaningful objects to gaze at sometimes. In a more realistic scenario, we believe that the termination of modulation works better by directing the user's attention to the vicinity of a more meaningful guidance area.

4 CONCLUSION

We proposed a subtle gaze guidance method for 360° content combining two techniques; gamma correction-based gradual brightness modulation and termination of modulation by gaze approaching the guidance area. Our method can be easily introduced to any VR content as post-processing without interfering with the natural viewing experience. Compared to existing methods, our modulation does not introduce unnatural color alteration and it is designed not to remain in the user's impressions. The experimental results confirmed that our method is less uncomfortable and it encourages natural viewing, whereas it yields comparable performance in terms of rough gaze guidance compared with more conspicuous methods.

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